

STRING FUNCTIONS IN MYSQL (WITH EXAMPLES)

1 Sample Table Structure

```
CREATE TABLE employees (  
emp_id INT PRIMARY KEY,  
full_name VARCHAR(100),  
email VARCHAR(100),  
department VARCHAR(50),  
city VARCHAR(50),  
salary VARCHAR(20),  
remarks VARCHAR(200)  
);
```

2 Insert Sample Data

```
INSERT INTO employees VALUES
```

```
(1, 'Karthik Kondpak', 'karthik.k@gmail.com', 'Data Engineering', 'Hyderabad',  
'75000', ' Top performer '),
```

```
(2, 'Veena Reddy', 'veena_r@company.com', 'Analytics', 'Bangalore', '65000',  
'Excellent communication'),
```

```
(3, 'Ravi kumar', 'ravi.kumar@org.in', 'Data Science', 'Chennai', '85000', 'Needs  
improvement'),
```

```
(4, 'Anil', 'anil@abc.com', 'DEVOPS', 'Pune', '70000', NULL),
```

```
(5, ' Suresh ', 'suresh@xyz.com', 'data engineering', ' hyderabad ', '60000', ' ');
```

1. LENGTH / CHAR_LENGTH

Returns number of characters

```
SELECT full_name, LENGTH(full_name) FROM employees;
```

```
SELECT full_name, CHAR_LENGTH(full_name) FROM employees;
```

Note: CHAR_LENGTH is safer for multi-byte characters.

2. UPPER / LOWER

Convert case

```
SELECT UPPER(department), LOWER(city) FROM employees;
```

3. TRIM / LTRIM / RTRIM

Remove spaces

```
SELECT
```

```
TRIM(full_name),
```

```
LTRIM(full_name),
```

```
RTRIM(full_name)
```

```
FROM employees;
```

4. CONCAT

Join strings

```
SELECT CONCAT(full_name, ' - ', department) AS emp_details
```

```
FROM employees;
```

5. CONCAT_WS (With Separator)

```
SELECT CONCAT_WS(' | ', emp_id, full_name, city)
```

```
FROM employees;
```

6. SUBSTRING / SUBSTR / MID

Extract part of string

```
SELECT SUBSTRING(email, 1, 7) FROM employees;
```

```
SELECT SUBSTR(email, 1, 5) FROM employees;
```

7. LEFT / RIGHT

Extract characters from left or right

```
SELECT LEFT(full_name, 4), RIGHT(city, 3)
```

```
FROM employees;
```

8. INSTR

Position of substring

```
SELECT email, INSTR(email, '@') FROM employees;
```

9. LOCATE

Same as INSTR but with more flexibility

```
SELECT email, LOCATE('.', email) FROM employees;
```

10. REPLACE

Replace substring

```
SELECT REPLACE(department, 'Data', 'Big Data')
```

```
FROM employees;
```

11. REVERSE

Reverse a string

```
SELECT full_name, REVERSE(full_name)
```

```
FROM employees;
```

12. LPAD / RPAD

Pad characters on left or right

```
SELECT LPAD(emp_id, 5, '0') FROM employees;
```

```
SELECT RPAD(city, 15, '*') FROM employees;
```

13. TRIM + REPLACE (Combined Usage)

```
SELECT TRIM(REPLACE(city, ' ', '')) FROM employees;
```

14. IFNULL

Handle NULL strings

```
SELECT full_name, IFNULL(remarks, 'No remarks')  
FROM employees;
```

15. COALESCE

Return first non-NULL value

```
SELECT full_name, COALESCE(remarks, 'N/A')  
FROM employees;
```

16. FIND_IN_SET

Search value in a comma-separated list (CSV)

```
SELECT FIND_IN_SET('Analytics', 'Data,Analytics,AI');
```